AQ9.6: Activity Questions 6 - Not Graded

This assignment will not be graded and is only for practice.

Level 1

1 point

Consider a function $f: \mathbb{R}^3 \to \mathbb{R}$.

Choose the correct statement with respect to ∇f.

- [ ] ∇f has co-domain $\mathbb{R}^2$
- [ ] ∇f has co-domain $\mathbb{R}$
- [ ] ∇f has co-domain $\mathbb{R}^3$
- [ ] ∇f has domain $\mathbb{R}^3$

No, the answer is incorrect.
Score: 0
Accepted Answers:
∇f has co-domain $\mathbb{R}^3$
∇f has domain $\mathbb{R}^3$

1 point

Let $f: \mathbb{R}^3 \to \mathbb{R}$ and suppose its gradient ∇f is continuous. Which of the following statements is true for the directional derivative of f in the direction of $a \in \mathbb{R}^3$? Consider dot product to be the inner product $<,>$.

- ( ) $D_a f = \frac{1}{\|a\|} < a, \nabla f >$
- ( ) $D_a f = < a, \nabla f >$
- ( ) $D_a f = \frac{1}{2\|a\|} < a, \nabla f >$
- ( ) $D_a f = \frac{1}{\|a\|^2} < a, \nabla f >$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$D_a f = \frac{1}{\|a\|} < a, \nabla f >$

Let $f(x, y) = e^x \sin y$. If ∇f at the point $(ln(2), \pi/4)$ is (k, k) where $k \in \mathbb{R}$, then find the value of k^2.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2

1 point

Level 2

1 point

Let $p = -\frac{k}{(\|r\|)^3}r$, where $r = (x, y, z) \in R^3 \setminus \{(0,0,0)\}, k \in R$.

Amongst the options below, choose those for which the function f satisfies $\nabla f = p$ for some $k \in R$.

Consider dot product as the inner product for finding the norm.

☐ $f(x, y, z) = \frac{1}{\|r\|}$

☐ $f(x, y, z) = \frac{1}{2\|r\|}$

☐ $f(x, y, z) = \frac{1}{\|r\|+2}$

☐ $f(x, y, z) = \|r\|$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$f(x, y, z) = \frac{1}{\|r\|}$

$f(x, y, z) = \frac{1}{2\|r\|}$

1 point

Let $f(x, y, z) = 4(x^2 + y^2) - z^2$.

Choose the correct option(s).

☐ ∇f at $(1, 0, 2)$ is $(8, 0, -4)$

☐ ∇f is continuous everywhere in R^3

☐ ∇f is not continuous everywhere in R^3

☐ ∇f at $(1, 0, 2)$ is $(-8, 0, -4)$

No, the answer is incorrect.
Score: 0
Accepted Answers:

∇f at $(1, 0, 2)$ is $(8, 0, -4)$

∇f is continuous everywhere in R^3

1 point

Consider a function defined as

$$f(x, y) = \begin{cases} \frac{y^3}{x^2 + y^2} & (x, y) \neq (0,0) \\ 0 & (x, y) = (0,0) \end{cases}$$

Choose the correct option(s).

☐ $\frac{\partial f}{\partial x} = \frac{-2xy^3}{(x^2 + y^2)^2}$, for $(x, y) \neq (0,0)$

☐
$\frac{\partial f}{\partial x} = \frac{2xy^3}{(x^2+y^2)^2}$, ∂x∂f = (x2+y2)22xy3, for $(x, y) \neq (0,0)$(x, y) = (0, 0)

☐
$\frac{\partial f}{\partial y} = \frac{3x^2y^2+y^4}{(x^2+y^2)^2}$∂y∂f = (x2+y2)23x2y2+y4 for $(x, y) \neq (0,0)$(x, y) = (0, 0)

☐
$\frac{\partial f}{\partial y} = \frac{3x^2y^2-y^4}{(x^2+y^2)^2}$∂y∂f = (x2+y2)23x2y2−y4 for $(x, y) \neq (0,0)$(x, y) = (0, 0)

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\frac{\partial f}{\partial x} = \frac{-2xy^3}{(x^2+y^2)^2}$, ∂x∂f = (x2+y2)2−2xy3, for $(x, y) \neq (0,0)$(x, y) = (0, 0)
$\frac{\partial f}{\partial y} = \frac{3x^2y^2+y^4}{(x^2+y^2)^2}$∂y∂f = (x2+y2)23x2y2+y4 for $(x, y) \neq (0,0)$(x, y) = (0, 0)

1 point
Consider the function definition given in Q6 above. Choose the correct option(s).

☐
$\frac{\partial f}{\partial x}$∂x∂f is continuous at $(0,0)$(0, 0)

☐
$\frac{\partial f}{\partial x}$∂x∂f is not continuous at $(0,0)$(0, 0)

☐
$\frac{\partial f}{\partial y}$∂y∂f is not continuous at $(0,0)$(0, 0)

☐
∇f∇f is discontinuous at the origin $(0,0)$(0, 0)

☐
Directional derivative of ff in the direction of a vector $a \in R^2$ a ∈ R2 at $(0,0)$(0, 0) can be obtained using the gradient of ff

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\frac{\partial f}{\partial x}$∂x∂f is not continuous at $(0,0)$(0, 0)
$\frac{\partial f}{\partial y}$∂y∂f is not continuous at $(0,0)$(0, 0)
∇f∇f is discontinuous at the origin $(0,0)$(0, 0)

1 point
Consider a function defined as

$$f(x, y) = \begin{cases} (x^2+y^2)\sin\left(\frac{1}{\sqrt{x^2+y^2}}\right) & (x, y) \neq (0,0) \\ 0 & (x, y) = (0,0) \end{cases}$$

f(x, y) = ⎩⎨⎧(x2 + y2) sin(x2+y2 √ 1)0(x, y) = (0, 0)(x, y) = (0, 0).

Choose the correct option(s).

☐
$\frac{\partial f}{\partial y}(0,0) = 0$∂y∂f(0, 0) = 0

☐
$\frac{\partial f}{\partial x}(0,0) = 0$∂x∂f(0, 0) = 0

☐
$\frac{\partial f}{\partial x} = 0$∂x∂f = 0

☐

$\frac{\partial f}{\partial y} = 0$

☐ None of the above

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\frac{\partial f}{\partial y}(0, 0) = 0$

$\frac{\partial f}{\partial x}(0, 0) = 0$

1 point

Consider the function definition given in Q8 above. Choose the correct option(s) with respect to the partial derivative of f. Note $sign(x) = \pm 1$ depending on the sign of x.

☐ $\frac{\partial f}{\partial x}(x, 0) = 2x\sin(\frac{1}{|x|}) - sign(x)\cos(\frac{1}{|x|})$ for $x \neq 0$

☐ $\frac{\partial f}{\partial x}(x, 0) = 2x\sin(\frac{1}{|x|}) - sign(x)\cos(\frac{1}{|x|})$

☐ $\frac{\partial f}{\partial x}(x, 0) = 2x\sin(\frac{1}{|x|}) - \cos(\frac{1}{|x|})$ for $x \neq 0$

☐ $\frac{\partial f}{\partial y}(0, y) = 2y\sin(\frac{1}{|y|}) - sign(y)\cos(\frac{1}{|y|})$ for $y \neq 0$

☐ $\frac{\partial f}{\partial y}(0, y) = 2y\sin(\frac{1}{|y|}) - sign(y)\cos(\frac{1}{|y|})$

☐ $\frac{\partial f}{\partial y}(0, y) = 2y\sin(\frac{1}{|y|}) - \cos(\frac{1}{|y|})$ for $y \neq 0$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\frac{\partial f}{\partial x}(x, 0) = 2x\sin(\frac{1}{|x|}) - sign(x)\cos(\frac{1}{|x|})$ for $x \neq 0$

$\frac{\partial f}{\partial y}(0, y) = 2y\sin(\frac{1}{|y|}) - sign(y)\cos(\frac{1}{|y|})$ for $y \neq 0$

1 point

Consider the function definition given in Q8 above. Choose the correct option.

☐ ∇f is continuous everywhere

☐ ∇f is discontinuous at the origin $(0, 0)$

☐ $\frac{\partial f}{\partial x}$ is continuous at the origin but $\frac{\partial f}{\partial y}$ is discontinuous at the origin

☐ $\frac{\partial f}{\partial y}$ is continuous at the origin but $\frac{\partial f}{\partial x}$ is discontinuous at the origin

☐ Directional derivative of f in the direction of a vector $a \in R^2$ at $(0, 0)$ cannot be obtained using the gradient of f

No, the answer is incorrect.
Score: 0

Accepted Answers:

∇f is discontinuous at the origin $(0, 0)$

Directional derivative of f in the direction of a vector $a \in R^2$ at $(0, 0)$ cannot be obtained using the gradient of f

Check Answers

Your score is: 0/10